AI-Driven Validation of IoT Data for Blockchain-Based Agricultural Supply Chain Traceability: A Survey of Oracle Mechanisms and AI Approaches

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Abstract: Blockchain-based traceability systems are often proposed as a solution to the opacity, fragmentation, and fraud that characterize agricultural supply chains. The property that these systems actually provide, however, is narrower than is usually claimed: a distributed ledger guarantees that a record, once committed, cannot be changed without detection, but says nothing of whether the record was true at the moment of commitment. In an agricultural supply chain, this distinction is crucial: nearly every on-chain fact originates from an IoT sensor in an uncontrolled environment or from manual entry by a party with an economic stake in the outcome. This survey reviews the literature addressing that gap. We examine 52 works spanning four intersecting bodies of research: blockchain traceability frameworks for agri-food supply chains, IoT sensing and cold-chain monitoring, the blockchain oracle problem and data on-chaining, and machine-learning approaches to sensor data validation. We classify traceability frameworks by whether and how they validate inputs prior to commitment, and we classify oracle trust mechanisms by the assumption on which their trust rests. Our principal finding is that the agri-food traceability literature and the oracle trust literature have developed largely in isolation: traceability frameworks overwhelmingly treat sensor readings as ground truth at ingestion, while oracle trust research is oriented toward financial data feeds and rarely addresses the physical, domain-constrained data characteristic of perishable logistics. We identify six research gaps arising from this separation, of which the most consequential is that no reviewed work clearly distinguishes implausible data, which should be rejected, from plausible data reporting a genuine violation, which must be recorded and attributed.

Index Terms: Blockchain, oracle problem, agricultural supply chain, traceability, Internet of Things, anomaly detection, smart contracts, data integrity, cold chain.

I. INTRODUCTION

Agricultural supply chains connect a large number of weakly interacting actors — farmers, aggregators, transporters, warehouses, retailers, and consumers — through long distances and extended time horizons. The resulting structure is characteristically opaque. Records are fragmented across the parties that create them, in incompatible formats, and in many regions on paper or not at all. The consequences are well documented: delayed product recalls, food adulteration and fraud, disputed liability for spoilage, and a persistent asymmetry of information in which the producer has the least visibility into the chain and captures the smallest share of the final value [1][3].

Blockchain technology has been proposed extensively as the structural answer to this opacity. A distributed ledger offers a shared record that no single participant controls, that all participants can read, and that none can retroactively alter. Smart contracts add programmable enforcement, allowing custody transfers, quality thresholds, and settlement conditions to execute without a trusted intermediary. A substantial body of work now demonstrates such systems for agri-food traceability, from early platform implementations [1] to crop-specific deployments [2] and provenance frameworks designed for scale.

The claim these systems make, however, is often overstated in a particular and critical way. Immutability is a property of the record, not the world described. A blockchain assures the immutability of a stored temperature reading but gives no assurances about the sensor which produced that reading. Was it operating correctly? Did it exist in the location where it transmitted the reading from? Was it tampered with before transmission began? Having committed a false value to the blockchain, immutability preserves it with exactly the same fidelity as a true value, and grants it the same aura of truth. The system is, exactly, tamper-evident but trust less.

It is a known issue in the blockchain literature which goes by the name of the oracle problem: smart contracts cannot natively access external state, and any mechanism which imports such state becomes a point at which the trust guarantees of the ledger terminate [4], [5], [8]. What is notable is how rarely the agri-food traceability literature engages with it. Caldarelli's examination of blockchain application papers found that only a small minority considered the role of oracles at all, and fewer still discussed the limitations oracles introduce [9]. Traceability

frameworks are typically evaluated on transaction latency, throughput, and gas cost; metrics which measure how efficiently the system records data, and say nothing about whether the recorded data deserved to be there.

The gap is less significant in other areas of application, where financial oracles operate on information that is already digital and subject to scrutiny from multiple sources: a spoofed price oracle could be detected by another exchange. A temperature sensor inside a refrigerated truck on a rural highway has neither a built-in redundancy nor an external reviewer. The information it provides is produced by an easily falsified technology, in a physically isolated environment, and in the custody of the entity that stands to benefit from the false information. The incentive to manipulate the data is direct, the opportunity to do so is unmonitored, and the ability to detect an attempted manipulation is limited.

This paper surveys the literature on methods proposed to address this gap, and identifies avenues of inquiry for future research. Specifically, we make the following contributions:

1) a survey of four areas of research that have seen only limited confluence to date: agri-food blockchain traceability, IoT cold chain sensing, oracle trust, and machine learning-based analysis of sensor streams;

2) a categorization of traceability systems according to how and whether they validate inputs submitted for inscription on-chain;

3) a categorization of oracle trust mechanisms according to their underlying trust assumptions, and an analysis of which mechanisms are suitable for processing physical sensors' data in time-critical settings;

4) and the identification of six research gaps, stemming from the limitations of each surveyed approach; in particular, we argue that an incorrect conflation of implausible sensor readings with authentic threshold-crossing events defeats the purpose of cold-chain monitoring.

The remainder of this article is structured as follows. Section II provides background information. Section III describes the methodology of this survey. Section IV presents the results of the survey in the form of a thematic literature review. Section V analyzes the findings in detail. Section VI identifies research gaps, while Section VII outlines challenges for future work. Finally, Section VIII provides a brief conclusion.

II. PROBLEM DOMAIN AND BACKGROUND

A. The Agricultural Supply Chain

A typical agri-food chain consists of a production phase, aggregation, processing, transport, storage, retail, and consumption phases, with custodians of the physical goods transferring at each boundary. Two features distinguish it from manufacturing supply chains. First, the product degrades continuously and irreversibly as a function of environmental exposure, so the conditions of transit are themselves part of product quality rather than incidental to it. Second, the upstream participants are numerous, small, and weakly organised, which means the party with the greatest interest in transparency is also the party least able to demand it.

B. Blockchain and Smart Contracts

A blockchain is an append-only ledger replicated across participating nodes, in which agreement on state is reached by a consensus protocol rather than by a central authority. Smart contracts are deterministic programs executed by every node, whose outputs are therefore agreed by consensus. Determinism is what makes consensus possible, and it is also the source of the oracle problem: a contract that could issue an external network request would produce different results on different nodes at different times, breaking agreement [5].

Permissioned ledgers such as Hyperledger Fabric and permissionless ones such as Ethereum have both been applied to traceability, with differing trade-offs in throughput, cost, and governance [1], [19]. The choice affects performance but not the oracle problem, which arises identically in both.

C. IoT Sensing in Perishable Logistics

Cold-chain instrumentation often consists of temperature and humidity sensing, location tracking, and occasionally shock or light detection, transmitted over cellular or short-range links to a gateway. The operating environment is hostile to measurement in mundane ways: sensors exhibit calibration drift over extended use, low-temperature operation increases measurement error, connectivity is intermittent on rural routes, and power constraints induce sparse sampling [26], [27]. These are ordinary engineering difficulties, but they matter here, because each produces data that is wrong without being malicious — and a validation mechanism must differentiate the two.

D. Blockchain Oracles and Data On-Chaining

An oracle is any mechanism that supplies external state to a smart contract. Oracles are commonly classified across multiple axes: inbound versus outbound, push versus pull, software versus hardware, and centralised versus decentralised [12], [13]. Heiss et al. reframed the problem as one of data on-chaining, arguing that the relevant question is not simply how data reaches the chain, but what trust properties the on-chaining pipeline preserves [6]. This framing is important for the present survey, because it directs attention to the pre-processing that occurs between sensor and ledger — precisely where validation would have to sit [7].

E. Anomaly Detection for Sensor Streams

Anomaly detection methods relevant to telemetry can be grouped into several classes. Statistical and threshold-based methods detect values lying beyond a fixed or adaptive threshold; these approaches are interpretable and cheap, but they are unable to detect anomalies with unremarkable values. Distance- and density-based techniques find points that lie distant from the core of the data distribution. Tree-based isolation techniques such as the isolation forest [16] utilize the

observation that anomalies tend to be isolated quicker by randomly partitioning the feature space, requiring no labels. Reconstruction-based deep methods, notably the LSTM encoder-decoder paradigm [17], learn the time structure and detect points that the model is unable to reconstruct, and are therefore suitable for detecting drift and flatline faults that are not apparent at a point-wise level. Pang et al. give a complete taxonomy of deep learning approaches [18].

III. SURVEY METHODOLOGY

A systematic database search was conducted across IEEE Xplore, ACM DL, ScienceDirect, SpringerLink, and Google Scholar, with inclusion of backward and forward snowballing from the main oracle surveys [4], [5].

The search strategy included combinations of terms related to each of the four areas of application, namely: ("blockchain oracle" OR "data on-chaining") AND (IoT OR sensor) AND trust; ("oracle problem") AND ("supply chain" OR traceability); ("anomaly detection") AND ("cold chain" OR perishable); (agriculture OR "agri-food") AND blockchain AND (IoT OR sensor).

The years included in the search were 2016 – 2026. The starting year was set to 2016 because the oracle problem has been recognized as a separate topic of research only recently – earlier works on algorithms for blockchain oracles were included as they appeared as precursors to the present body of research.

Inclusion criteria were: published in a refereed journal or preprint repository, written in English, addressed blockchain-based traceability, oracle on-chaining trust, or sensor anomaly detection, described a method, an architecture, or presented an evaluation.

Exclusion criteria were: no significant methodological contribution, published in a non-research venue, duplicates, or available only as a commercial white paper with no independent evaluation.

The screening process was carried out in three stages. Firstly, 214 papers were retrieved from the databases. After duplicates were removed, 176 papers were subjected to title and abstract screening. Following that, 78 papers were included for full-text review. Finally, 52 papers were included in the review, of which 21 are discussed in detail in Section V of this article comparing the approaches.

For each paper, the application area, source of data, if inputs were validated prior to being committed to the blockchain, the method or algorithm used, evaluation data or results, reported results, and limitations, if any, were extracted. This information is presented in the comparative tables of Section V.

IV. LITERATURE REVIEW

A. Blockchain Foundations for Agri-Food Traceability

The foundational implementations established that end-to-end traceability on a distributed ledger is technically feasible. Caro et al. presented AgriBlockIoT, a decentralised traceability system that integrates IoT devices with the ledger, and notably implemented it on both Ethereum and Hyperledger Sawtooth to permit direct comparison of latency, CPU, and network cost [1]. Salah et al. demonstrated a crop-specific Ethereum implementation for soybean traceability, using smart contracts to govern transitions between supply-chain stages [2]. Biswas et al. applied similar principles to wine, where provenance carries a direct price premium and counterfeiting is economically attractive [22]. Malik et al. addressed the scalability limits these systems encounter, proposing a layered framework to support provenance queries at volume [21].

Broader reviews confirm both the breadth of activity and its consistent shape. Casino et al., surveying blockchain applications across domains, found supply chain to be among the most active application areas while noting a prevalence of proof-of-concept evaluation over deployment [3]. Panarello et al. surveyed blockchain–IoT integration specifically, identifying device constraints and data volume as recurring obstacles [19].

What is consistent across this body of work is the treatment of the input. Data arriving from an IoT device is written to the ledger, and the contribution of the system is located in what happens afterwards — how efficiently it is stored, how it is queried, who may read it. The question of whether the arriving data was correct is, with few exceptions, outside the scope of these frameworks. This is not an oversight on the part of individual authors so much as a shared framing, and it is the framing this survey seeks to make visible.

B. IoT Integration and Cold-Chain Monitoring

A parallel literature addresses the sensing layer on its own terms. Work on cold-chain monitoring establishes the practical realities of the measurement environment. Studies of anomaly detection in cold-chain data streams identify strong noise, elevated measurement error at low temperature, accuracy degradation over sensor lifetime, and high data volume and dimensionality as the defining challenges [26]. Deployment-oriented work reports systems built on business-defined thresholds rather than learned models, with decision support delivered by alerting [27] — an approach that is robust and interpretable but incapable of detecting faults whose values fall within nominal range.

Vangala et al. examined secure sensing for agricultural IoT from a blockchain perspective, addressing device authentication and the security of the sensing layer itself [20]. This is an important complement to the traceability literature, because it targets the assumption that the traceability frameworks leave implicit: that the device reporting is the device it claims to be.

The cold-chain and traceability literatures, however, rarely cite one another. Work on sensor reliability tends to terminate at the alerting or dashboard layer; work on blockchain

traceability tends to begin at the point where a reading has already been accepted.

C. The Oracle Problem and Data On-Chaining

The oracle literature has much to say about trust boundaries. Egberts was among the first to identify the problem, arguing that trusting external information sources nullifies the decentralization advantages of using a ledger in the first place [8]. Al-Breiki et al. provided the canonical treatment of the trust in oracles problem, analyzing the notion of trust itself and comparing trust-enabling characteristics of different approaches and platforms [4]. Pasdar et al. provided a survey of oracle implementations, framing the trust question in terms of connecting programmable contracts to the systems storing state information [5], and identifying common design patterns in oracle design [13].

A number of common approaches to building trust in oracles have been suggested. Trusted execution environments allow for self-reporting, attesting that a certain piece of information came from a particular source without any modifications [24]. Decentralized networks implement multiparty reporting and aggregate results, replacing trust in a single entity with trust in the majority being correct. Reputation systems introduce weights to different reports according to the reporters' known reliability [14]. More recently, machine learning techniques have been applied to the oracle problem: reinforcement learning allows for finding optimal balances between trust and cost [14], [25].

Heiss et al.'s work provides the closest publicly available treatment of the question. Their analysis of the data on-chaining paradigm provides the most detailed look at the integrity guarantees of a system with a data pre-processing stage. By introducing such a stage, the trust assurances of the system can no longer be absolute – while the signature over raw sensor data can be verified against the smart contract's address, the same cannot be said of any post-processing stages. The authors note that this creates a trust assurance gap, providing an opportunity for malicious actors to exploit the system [7]. This seems to be the most similar analysis to the problem identified in this paper: the agricultural use case described here shares its premise of a decentralized trust assurance system with weak integrity guarantees.

Caldarelli et al. describe a similar trust assurance problem, but in the domain of traceability rather than raw sensor data [9]. They note that tracking physical objects presents unique challenges, where the uniqueness of every item prevents simple cryptographic solutions. This adds to the list of areas where conventional blockchain-based approaches fail, requiring additional trust-assurance mechanisms. Al Sadawi et al. provide a survey of the blockchain-IoT-oracle space, identifying similar challenges [10]. Albizri and Appelbaum discuss the paradigm shift that led to the development of blockchains and oracles, and how it created new trust relationships in the system [11].

The difference between the identified issue and the existing literature is the domain in which it emerges. The vast majority of oracle research is conducted in the trustless finance space, where the value of information is directly linked to financial gain. This creates strong economic incentives for data manipulation, and the data itself is often discrete, binary, and easy to verify. Sensor data, on the other hand, is continuous and subject to physical laws – there is a concept of a valid interval rather than a specific value. The data has an inherent trust root in the sensor itself, although that trust can be undermined by analysis of the equipment. While financial data can be manipulated for direct financial gain, sensor data's value is indirect – in liabilities and potential for fraud.

D. Machine Learning for Sensor Data Validation

Work combining machine learning with blockchain for IoT data is growing greatly, though the focus is more on intrusion detection than data validation. Surveys of IoT anomaly detection catalog the available techniques and note blockchain as an emerging complement, mainly for tamper-evident logging of detection outcomes [28], [29]. Several architectures put deep anomaly detection at the edge and commit the results to a ledger, which provides auditability for the detection process rather than the underlying stream.

For the algorithms themselves, the relevant families are well established. Isolation Forest allows unsupervised detection with low computational cost and does not require labelling [16]. LSTM encoder-decoder architectures model temporal dependency and detect anomalies through reconstruction error, tackling fault classes like drift and flatline that pointwise methods miss [17]. Pang et al. survey the deep learning landscape comprehensively [18].

A few works apply such methods explicitly at the ingestion boundary. Lv et al. tackled location spoofing detection in blockchain-based IoT systems using a fuzzy analytic hierarchy process to assess the trustworthiness of location proofs [23] — a direct example of validating a specific claim class before accepting it. This remains the exception rather than the pattern.

E. Device-Level and Cryptographic Integrity

Finally, a strand of research focuses on integrity at the device. Authentication schemes tie readings to a device identity, and hardware mechanisms such as secure elements and physical unclonable functions aim to make that identity hard to extract. Al Breiki et al. [15] combined blockchain with trusted oracles for decentralised access control over IoT data. These mechanisms are necessary and complementary but insufficient on their own: cryptographic authentication establishes that a reading originated from a specific device, but says nothing about that device's veracity. A compromised or inaccurate sensor produces readings that are, cryptographically, the same as any other.

TABLE I. Comparative Analysis of Reviewed Traceability and Validation Frameworks

Ref	Domain / Focus	Data Source	Input Validation Before Commitment	Method / Platform	Evaluation Basis	Stated Limitation
[1]	Agri-food traceability	IoT devices	None stated	Ethereum and Hyperledger Sawtooth	Latency, CPU, network usage	Preliminary evaluation; no input trust model
[2]	Soybean supply chain	Actor-entered	None stated	Ethereum smart contracts	Functional demonstration; cost	Assumes honest actor input
[21]	Supply-chain provenance	Actor-entered	None stated	Layered permissioned framework	Scalability of provenance queries	Input integrity out of scope
[22]	Wine provenance	Actor-entered	None stated	Blockchain traceability system	System demonstration	Physical–digital binding unaddressed
[20]	Agricultural IoT sensing	Sensor devices	Device authentication	Blockchain-based secure sensing	Security analysis	Authenticates device, not reading veracity
[7]	Generic IoT on-chaining	Sensor devices	Yes — verifiable pre-processing	Trustworthy pre-processing workflow	Security argument; prototype	Not domain-specific; no policy semantics
[23]	IoT location proofs	Location data	Yes — spoofing detection	Fuzzy AHP evaluation	Simulation	Single claim class (location only)
[26]	Cold-chain logistics	Sensor stream	Detection only, off-chain	Stream anomaly detection	Detection accuracy	No ledger integration
[27]	Cold-chain transport	Sensor stream	Threshold rules only	IoT alerting and dashboard	Operational deployment	Cannot detect in-range faults
[14]	Industrial IoT oracle	Heterogeneous off-chain	Yes — node selection and reputation	Oracle scheme with QoS balancing	Security and QoS analysis	Not applied to physical perishables

TABLE II. Classification of Oracle Trust Mechanisms by Underlying Trust Assumption

Mechanism Class	Representative Work	Trust Assumption	Strength	Limitation for Perishable Logistics
Trusted execution environment	[24]	Hardware enclave is not compromised	Attests data was retrieved unmodified from a named source	Attests transmission fidelity, not sensor truthfulness
Decentralised oracle network	[4], [5]	A majority of independent reporters are honest	Removes single point of trust	Requires multiple independent observers; a single truck has one sensor
Reputation-weighted	[14]	Past reliability predicts future reliability	Degrades gracefully; no extra hardware	Slow to react to sudden compromise or calibration drift
Cryptographic device identity	[15], [20]	Device key material is not extractable	Establishes provenance of the reading	Authenticated falsehoods remain authenticated
ML-based validation	[16], [17], [23]	Normal behaviour is learnable and stable	Detects faults that are individually in-range	Requires labelled or clean training data; threshold selection is arbitrary without benchmarks
Verifiable pre-processing	[7]	Processing step can be independently verified	Restores end-to-end integrity across transformation	Addresses transformation integrity, not semantic plausibility

V. COMPARATIVE ANALYSIS

Tables I and II present the comparative synthesis. Table I classifies the reviewed frameworks according to the dimension this survey argues is decisive, namely, whether inputs are validated before they are committed. Table II classifies oracle trust mechanisms according to the assumption on which their trust ultimately rests and assesses them against the specific conditions of perishable logistics.

Three observations follow from the comparison.

First, the 'input validity' column of Table I is overwhelmingly empty. Of the reviewed agri-food traceability frameworks, validation is either absent or restricted to fixed thresholds; where it appears in a substantive form, it comes from outside the agricultural literature entirely --- from generic IoT on-chaining research [7] or from work on a single narrow class of claims such as location [23]. The systems designed for the domain do not validate; the systems that validate are not designed for the domain.

Second, the evaluation bases are not commensurable. Frameworks report latency, gas cost, CPU utilisation, and throughput; detection systems report precision, recall, and accuracy on datasets that are private, synthetic, or drawn from unrelated domains. No two of the reviewed works are evaluated on a shared benchmark. Consequently, claims of superiority within either literature cannot be compared across it, and a practitioner choosing between approaches has no empirical basis for doing so.

Third, Table II exposes a structural mismatch between available oracle mechanisms and the conditions of agricultural logistics. Decentralised oracle networks, the dominant answer to oracle trust in financial contexts, presuppose multiple independent observers of the same fact. A consignment in transit has one sensor. Redundant instrumentation is conceivable but the marginal cost falls on precisely the participants least able to bear it. Trusted execution environments and cryptographic device identity address the integrity of transmission and the authenticity of origin, but neither speaks to the semantic question of whether a well-formed, correctly signed, faithfully transmitted reading is actually true.

Taken together, these observations indicate that the trust problem in agricultural traceability is not merely under-solved but under-posed. The available mechanisms answer questions about provenance and transmission; the question that determines whether the ledger is worth trusting is one of plausibility.

VI. RESEARCH GAPS

We identified six gaps from the preceding analysis.

Gap 1: Input validation is assumed rather than applied. The vast majority of traceability frameworks reviewed treat sensor readings as immutable upon ingest [1], [2], [21], [22]. Immutability is applied to unvalidated data, creating a permanent record of possibly erroneous data. None of the reviewed agri-food frameworks incorporate a validation stage at ingest as an architectural feature.

Gap 2: The oracle and agri-food literatures touch on related concerns but intersect rarely. Oracle trust research is focused on financial feeds and rarely touches on the physical, domain-restricted nature of sensing [4], [5], [8]. On the other hand, agri-food traceability work rarely references the oracle literature at all. The two bodies of work are addressing a common underlying issue from opposite directions and have not yet intersected in practice.

Gap 3: Anomaly and violation are conflated. This is by far the most significant gap identified. An anomalous reading — a location jump of hundreds of kilometres within seconds, a temperature rise of forty degrees Celsius and return within one sampling interval — is data that should not be trusted. A threshold violation — a measured, legitimate excursion beyond the safe range for a crop — is data that must be taken as true and logged appropriately to whoever held custody. Both present as outlying temperature values with no additional information. None of the reviewed work treats them as different classes of information that should be handled in fundamentally different ways. A system that classifies threshold violations as anomalies and discards them would be actively suppressing precisely the events that cold-chain monitoring is intended to capture.

Gap 4: There exists no shared benchmark or labelled dataset. Evaluation of sensor-data trust in cold chain is performed on private, proprietary, or synthetic data without published ground truth [26], [27]. Reported detection accuracies are thus not comparable between works, and there is no baseline from which a new method can be benchmarked.

Gap 5: There exists no economic transparency. Traceability frameworks log the physical provenance of produce — where it originated, what path it followed, and under what conditions. Almost none log the price at which custody was transferred at each stage. Consequently, the systems can establish provenance but not the distributional question that motivates much agricultural policy interest in transparency: what share of final retail value finds its way to the producer.

Gap 6: Evaluation is dominated by proof of concept implementations. Consistent with Casino et al.'s [3] observation, most reviewed implementations report functional demonstrations and few report performance under sustained load, and even fewer report gas cost, write latency, or throughput at realistic telemetry volumes. Claims of scalability are therefore largely unsubstantiated.

VII. CHALLENGES AND LIMITATIONS

There are several challenges which explain why the gaps above persist and constrain the ease with which they can be closed.

Measurement environment. Sensors in cold-chain deployment drift, report higher error at low temperature and degrade in accuracy over service life [26]. Any validation mechanism must distinguish such benign failure modes from

adversarial manipulation, and the signatures are not easily disentangled.

Economics of instrumentation. Redundant sensing would resolve most of the trust problem by providing the independent observers that decentralised oracle designs assume. The cost is borne by smallholder producers and small logistics operators, for whom it is prohibitive. This is a structural rather than technical barrier.

Cost and latency of on-chain writes. High-frequency telemetry is incompatible with per-reading on-chain commitment at public network prices. Hybrid designs committing hashes or aggregates reduce the cost but weaken the granularity of what is actually attested.

Privacy against transparency. Transparency is the stated objective, but commercial pricing, supplier relationships and personal data of participants are sensitive. Data protection regimes which confer erasure rights are hard to reconcile with immutable storage, which constrains what may legitimately be placed on-chain.

Absence of labelled anomaly data. Supervised approaches are largely ruled out in this domain because genuine fault events are rare, unlabelled and commercially sensitive when they occur.

Limitations of this survey. Our search was limited to five databases and to English-language publications, and the four-area scope makes depth trade off against breadth in each. The selection of the comparative subset involved judgement about representativeness which another reviewer might have exercised differently. Finally, the oracle literature is proliferating rapidly, and recent contributions may not have found their way into the citation networks used for snowballing.

VIII. FUTURE SCOPE

The gaps identified suggest directions for research.

The most immediate is the architectural separation of plausibility assessment from policy evaluation. A validation layer between acquisition and commitment could assess whether a reading is physically credible — using rate-of-change constraints, implied-velocity checks, and cross-sensor consistency — and independently assess whether a credible reading violates a crop-specific threshold. The former determines whether data is committed; the latter determines what is recorded alongside it. Separating the two resolves Gap 3 directly and is implementable with existing methods.

A second direction is the construction of a public benchmark for cold-chain sensor trust, comprising realistic telemetry with injected faults of known type, timing, and magnitude. Ground truth recorded at injection time would permit precision and recall to be computed and compared across methods, addressing Gap 4. The absence of such a resource is arguably the single largest impediment to progress in this area.

Third, the deliberate transfer of oracle trust mechanisms into the agricultural setting merits investigation. Reputation-weighted schemes [14] map plausibly onto devices with service histories; verifiable pre-processing [7] maps onto the validation layer described above. Whether the assumptions these mechanisms rely upon survive transfer to a domain with single observers and physically constrained data is an open empirical question.

Fourth, extending traceability data models to record transfer price alongside transfer of custody would enable analysis of value distribution across the chain, addressing Gap 5 at negligible technical cost. The obstacles here are commercial and governance-related rather than technical.

Finally, systematic reporting of gas consumption, write latency, and throughput under sustained telemetry load would allow the scalability claims common in this literature to be assessed rather than assumed.

IX. CONCLUSION

This survey investigates the trust status of data entering blockchain-based agricultural traceability systems, synthesizing literature on agri-food traceability, cold-chain IoT sensing, blockchain oracle trust, and machine-learning anomaly detection. Its main finding is one of separation: that the literature designing traceability systems for agriculture does not validate its inputs, and the literature addressing input trust is oriented toward a domain with different data characteristics and different adversarial structure. The consequence is a class of systems that are cryptographically robust and epistemically fragile — able to prove that a record has not changed, but unable to establish that it was ever correct.

Of the six gaps identified, the conflation of implausible data with genuine threshold violations is the most consequential, because it is a design error whose effect is to suppress exactly the events the system exists to detect. Resolving it requires no new algorithm, only the recognition that the two categories demand opposite handling.

We further find that the absence of a shared benchmark prevents the field from establishing whether progress is being made at all, and that the omission of price from traceability data models forecloses the distributional analysis that motivates much of the policy interest in agricultural transparency. Addressing these would substantially strengthen the case that blockchain traceability delivers the trust it is widely claimed to provide.

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